

Course Project: Code Checker

Theodore Skafidas

Bench #28

EECE.3110 Electronics I

Date Submitted: 12/11/24

Due Date: 12/11/24

I. Summary

The purpose of this project is to apply learned knowledge of electronic components and problem-solving skills to create a circuit that reflects course content.

II. Equipment

Table I. Equipment Used

Equipment Type	Details
Bench “shoebox”	N/A
Dual Power Supply (+12V & -12V)	Make: GW Instek Model: GPD – 3303D Serial #: EL832093
DMM Bench-top	Make: Keithley Model: 2110 5.5 Digit Multimeter Serial #: 8006828

Table II. Components Used

Component Type	Details
Resistor	1k ohm
Resistor	4k ohm
LED	Green/Red
BJT	npn
Op-Amp	741

III. Introduction

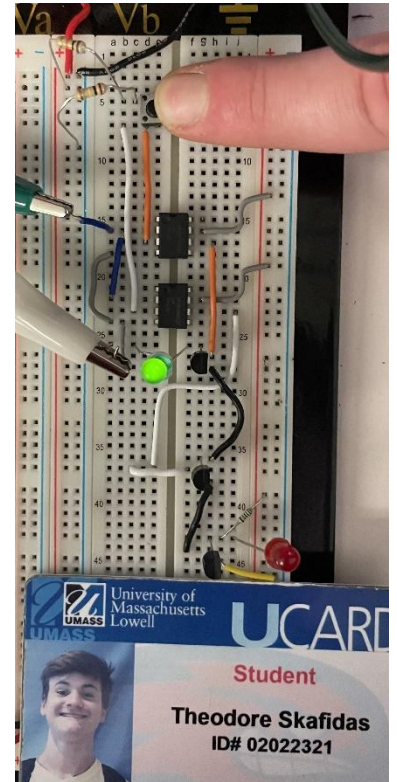
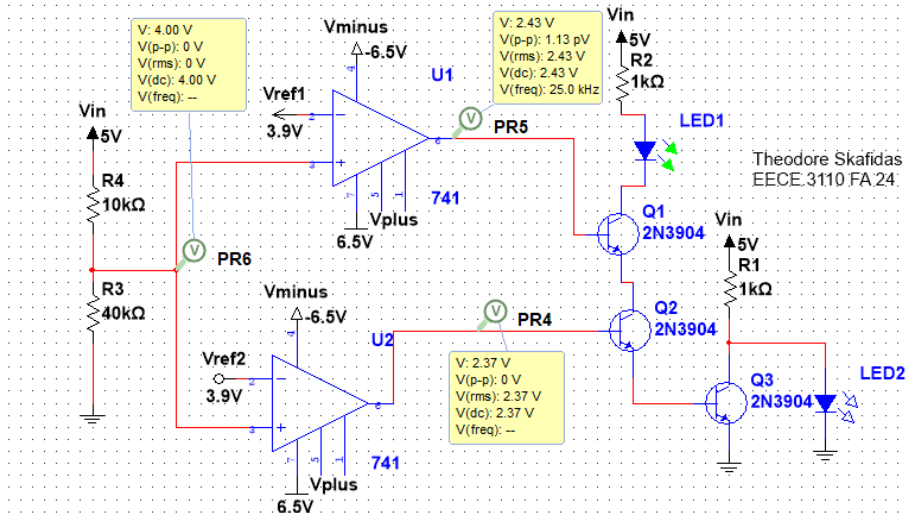
The design of the Code Checker is to confirm the values and the order of two resistors and to create a signal to indicate the correct ‘code’. The ‘code’ is the combination of 2 resistors. The ‘checker’ will check the values and order of the resistances using operational amplifiers. The correct condition of values and order will be controlled by a combination of transistors. The correct condition is indicated by a green LED, with all other conditions indicated by a red LED. If the incorrect combination of resistors is present, the red LED will be on, and the green LED turned off. If the correct combination is met, the circuit will turn on the green LED and turn the red LED off.

Voltage Divider

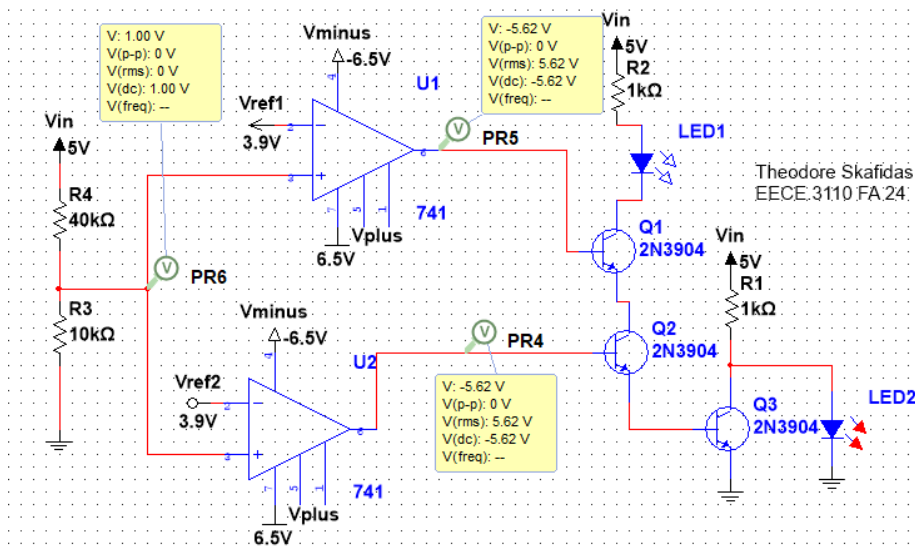
$$V_{ref} < \frac{R_4}{R_3 + R_4} * V_{in}$$

(Formula used to calculate V_{ref})

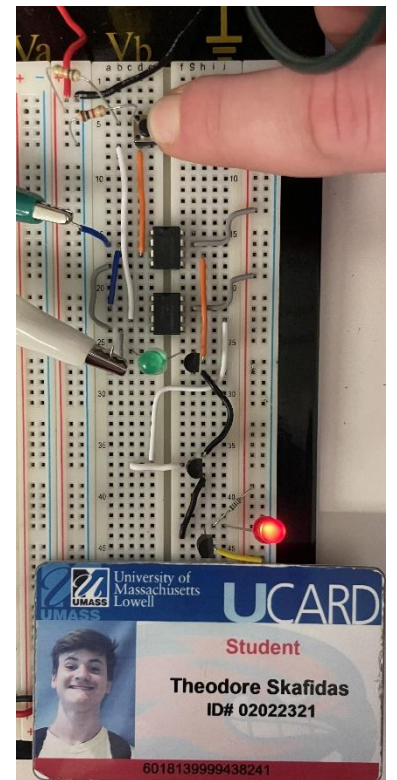
IV. Circuit Description



Code Checker circuit with the correct resistor combination



Code Checker circuit with an incorrect combination of resistors



V. Measurements

The below measurements include calculations for V_{ref} values used in the circuit.

$$V_{ref} < \frac{40k}{10k + 40k} * 5 = 4V$$
$$V_{ref} \cong 3.9V$$

VI. Discussion

This design was achieved using operational amplifiers and npn-type bipolar junction transistors. The two ‘code’ resistors were in series from an input (V_{in}) to ground. A wire was connected in parallel with the resistors and fed into the non-inverting input of two op-amps.

Two op-amps ensure the correct order of the resistors is picked. The inverting input of the op-amps are connected to a voltage input (V_{ref}) equal to less than the output voltage (using voltage divider rule) of the two correct resistors. The op-amps are used as comparators, comparing the voltage of the inverting input (correct resistor combination) to the non-inverting input (selected resistor combination). When the inverting input is greater than the non-inverting input, the op-amp outputs a positive voltage; and a negative voltage if the inverse is true. This was confirmed in the MULTISM simulation.

The op-amps output is connected to the base of two BJTs in series. Above the BJTs is a green LED with an input voltage (V_{in}). The BJT’s act as a two-input AND gate. When a positive output voltage of both op-amps is present, the BJT’s allow current to flow from the collectors to the emitters, and from the green LED to ground. Otherwise, the gates will be closed and the green LED turned off. This was confirmed in the MULTISM simulation.

Past the AND gate, a wire is fed into the base of a third BJT. A red LED is connected to the third BJT’s collector. This BJT acts as a NOT gate. When current flows into its base, the gate will open

and allow current to flow away from the red LED, from the emitter to ground, turning the red LED off. This allows for the green LED's behavior to influence when the red LED is turned on and off.

The final circuit did have one flaw: it checked the ratio of the two resistors rather than the values of them. Both op-amps in the circuit reference the R3 resistor as a comparison to V_{ref} instead of one referencing R4 and the other referencing R3. This was the reason for equal V_{ref} values and because of this, the inclusion of a second op-amp was redundant. The current Code Checker circuit will work as long as the resistance of R3 is four times greater than the resistance of R4. While that wasn't the intention of the circuit, the error only affected the op-amps. The remaining portion of the circuit, including the BJTs connected to LEDs, emulated its intended purpose of LED signal control. With some design adjustments to the op-amps' use, the circuit could function as intended.

VII. Conclusion

This lab reinforced the application of op-amps and bipolar junction transistors in circuit design. The skills and knowledge gained from this project furthered my understanding of the behavior of important electrical components which will be used in future courses.

References

Bowden, D. A. (2024). EECE.3110 Course Project Fall 2024. University of Massachusetts, Lowell, Lowell, MA.

Bowden, D. A. (2019). Electrical and Computer Engineering Laboratory Requirements and Safety Manual. University of Massachusetts, Lowell, Lowell, MA.